🟡 Supported Models

Qwen3
4B / 8B / 32B

Qwen3-VL
4B / 8B

Qwen2.5
32B

DeepSeek-R1
Distill variants

GLM4
9B / 30B / 355B

⬛ Entry Points

🔧 **Terminal Environment**

🧱 **Math / Tool-Call Python Sandbox**

🖥️ **GUI / Desktop Environment**

🖥️ **SWE-Bench GitHub Repo**

👤 **User (OpenClaw App)**

bash tasks

math problems

screen / state

issue + repo

multi-turn conversation

🟩 TRACK 2 — General Agentic RL

Terminal Agent

🐳 Docker Pool Server
(pool_server.py :5000)

Container
SETA Tasks
(bash commands)

Tool-Call / Math

📐 ReTool Dataset
DAPO-Math-17k
AIME eval

🐍 Python Sandbox
(tool_sandbox.py)

GUI Agent

☁️ VM Pool Server
(env_pool_server.py)

Cloud VM
OSWorld Env
(pyautogui)

screenshot

🖼️ Qwen3-VL
Screenshot → Action

SWE-Bench Agent

🏗️ Env Pool Server
(:18090) + ECS nodes
(:5000)

Docker Container
GitHub Repo
Test Suite

reward signal

action + reward

patch success/fail

code output

⬛ Shared Infrastructure

🟣 SLIME Framework
(THUDM)

📦 Rollout Buffer
(Ray actors)
async trajectory queue

🟦 TRACK 1 — Personal

Training Methods

📊 Binary RL
(GRPO)
reward: +1/−1/0

📚 On-Policy
Distillation (OPD)
token-level / top-K

🔀 Combined
w_rl × GRPO +
w_opd × teacher

GRPO advantage

teacher log-probs

combined advantage

Deployment

🖥️ Local GPU
Full Fine-Tune
8+ GPUs

🔬 Local GPU
LoRA
1–2 GPUs

☁️ Tinker Cloud
No local GPU
LoRA only

API Layer

🌐 API Server
(FastAPI + SGLang)
OpenAI-compatible proxy

trajectories

serve response

⚡ SGLang Engine
(Policy Model Serving)
TP=2, context=32k

🎯 Reward Aggregator
score / filter
balance-data

scored trajectories

terminal reward

🟠 PRM / Judge Layer

🧠 Process Reward Model
(SGLang engine)
majority vote × m

⚖️ Outcome Judge
binary / hindsight
hint extractor

step-wise rewards

episode reward

model weights

gradient updates

🔴 Policy Trainer — Megatron-

📉 GRPO Loss
PPO clip [0.2, 0.28]
KL regularization

🔧 Adam Optimizer
CPU offload
precision-aware

⚙️ Megatron-LM
TP / PP / EP
distributed training

💾 Checkpoint Management

🤗 HuggingFace
checkpoint
(input / output)

🔄 HF ↔ Megatron
bridge converter

🗄️ Megatron
distributed ckpt
(sharded TP/PP)

load / save

📡 Observability

📊 Weights & Biases
loss / reward / eval

🖥️ Ray Dashboard
:8265